UNSUPERVISED PORTFOLIO OPTIMIZATION

A K-Means Clustering Strategy on S&P 500 Stocks

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ABSTRACT:
This project presents an unsupervised machine learning–based trading strategy that employs the K-Means clustering algorithm on S&P 500 stock data to identify hidden market structures and behavioral patterns. Rather than relying on predictive models, the approach emphasizes pattern recognition and portfolio diversification by grouping stocks with similar historical characteristics, including volatility, momentum, and beta. Clusters representing distinct risk–return profiles are analyzed to construct optimized portfolios using the Efficient Frontier framework, aiming to maximize the Sharpe ratio and achieve an optimal balance between return and volatility. The performance of these cluster-based portfolios is benchmarked against the S&P 500 index to evaluate their effectiveness in enhancing risk-adjusted returns. Overall, this study demonstrates how integrating unsupervised learning techniques with modern portfolio theory can yield adaptive, interpretable, and systematically diversified trading strategies in dynamic financial markets.

METHODOLOGY
This study follows a structured pipeline integrating data processing, clustering, portfolio optimization, and performance evaluation:

Data Collection & Feature Engineering:
S&P 500 stock data was collected using yfinance, and key financial features -volatility, momentum, beta, moving averages, and Sharpe ratio were engineered and aggregated on a monthly basis.

Clustering (K-Means):
The K-Means algorithm was applied monthly to the 150 most liquid stocks, grouping them based on feature similarity to identify distinct market behavior clusters.

Portfolio Optimization (Efficient Frontier):
Using Modern Portfolio Theory, the Efficient Frontier was computed, and the portfolio corresponding to the maximum Sharpe ratio was selected for optimal risk–return trade-off.

Backtesting & Evaluation:
The optimized portfolios were backtested over time, and performance metrics and cumulative returns were compared against the S&P 500 benchmark to assess strategy effectiveness.

KEY RESULTS:
This section highlights the major findings of the study through visual and quantitative analyses:

- Clustering Visualization:

(Insert 2D PCA plot)

A 2D projection of the feature space (via PCA) clearly distinguishes multiple stock clusters, each representing unique market behavior profiles—ranging from high-risk/high-return groups to more stable, low-volatility clusters.

- Efficient Frontier Plot:

(Insert Efficient Frontier chart)

The Efficient Frontier demonstrates the trade-off between risk and return, with the Maximum Sharpe Ratio portfolio visibly marked as the optimal point, representing superior risk-adjusted performance.

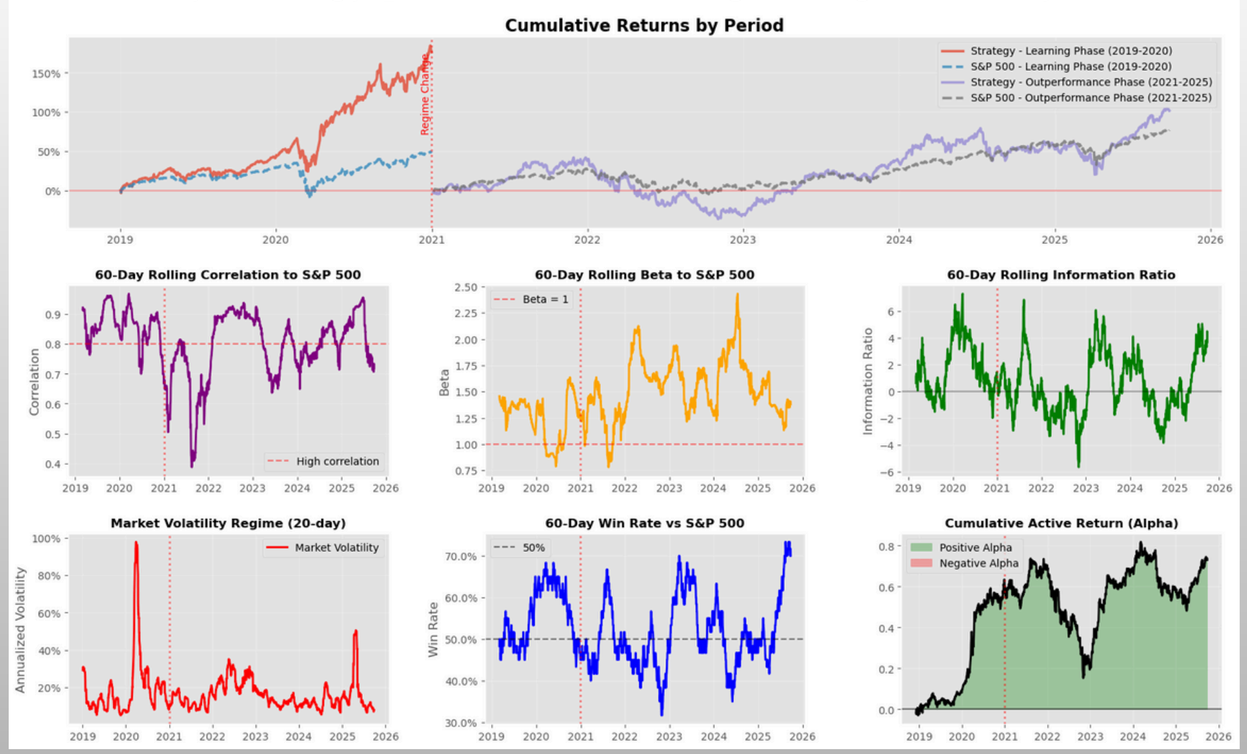
- Backtesting Performance:

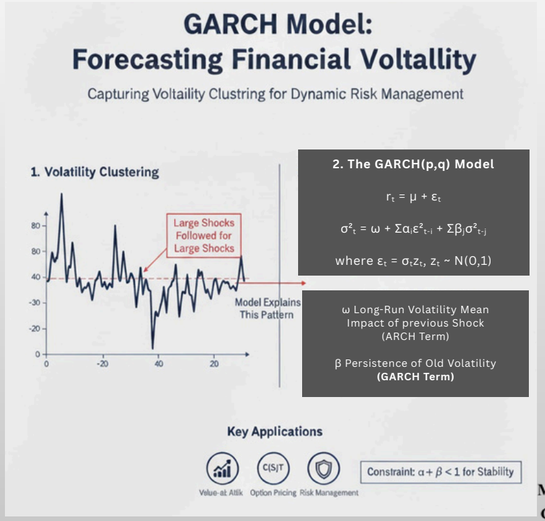
(Insert Cumulative Returns line chart)

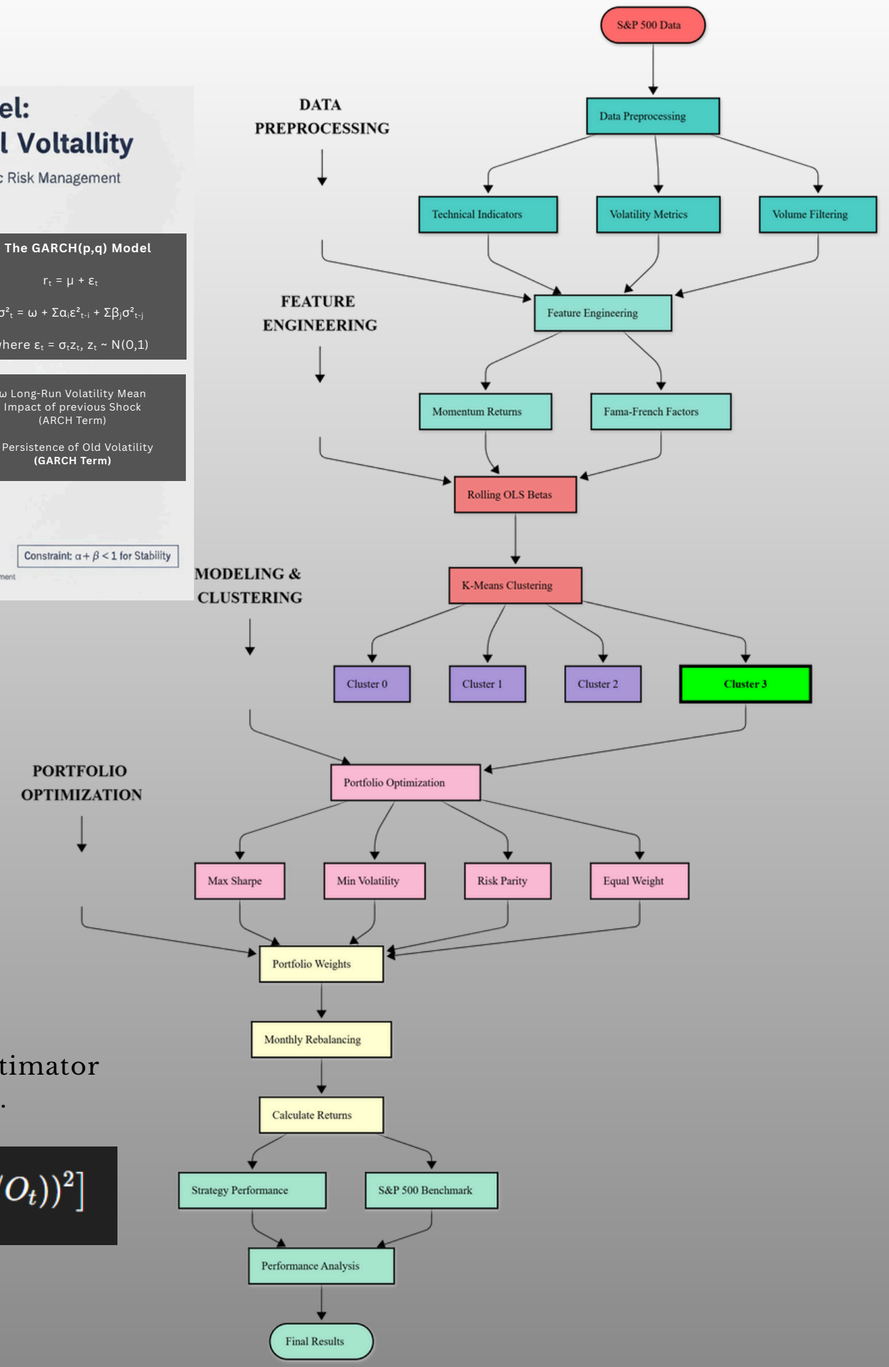
Backtesting results show that the Clustered Portfolio achieves improved diversification and more stable returns compared to the S&P 500 benchmark, often exhibiting lower drawdowns and enhanced consistency, even when not always outperforming the index.

INTERPRETATION:
The clustering approach effectively groups stocks with similar financial characteristics, enabling more balanced portfolio construction. Optimization using the Efficient Frontier enhances risk-adjusted returns, demonstrating that an unsupervised, data-driven framework can provide stability and diversification benefits within dynamic market conditions.

Why strategy performance changed Regime Analysis?







Volatility (Feature Engineering):

$$\text{Volatility}(\sigma) = \sqrt{\frac{\sum_{i=1}^{N}(R_i - \bar{R})^2}{N-1}}$$

Momentum (Feature Engineering):

$$\text{Momentum} = \frac{P_t}{P_{t-n}} - 1$$

Sharpe Ratio (Portfolio Optimization Goal):

$$\text{Sharpe Ratio} = \frac{(R_p - R_f)}{\sigma_p}$$

Fama-French 3-Factor Model regression equation:

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_i(R_{m,t} - R_{f,t}) + s_i\text{SMB}_t + h_i\text{HML}_t + \epsilon_{i,t}$$

Garman–Klass volatility, a well-known range-based estimator of volatility that uses high, low, open, and close prices.
Garman–Klass estimator of variance is:

$$\sigma^2_{GK} = \frac{1}{n}\sum_{t=1}^{n}\left[0.5(\ln(H_t/L_t))^2 - (2\ln 2 - 1)(\ln(C_t/O_t))^2\right]$$

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